Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)

HEALTHY HARVEST

Submitted in partial fulfillment of the requirement
of the degree of

Bachelor of Technology in
Information Technology

By

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Under the guidance of

Prof. Arjun Jaiswal

University of Mumbai

A.Y. 2022 – 2023

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)

CERTIFICATE

This is to certify that the project entitled, **"Healthy Harvest"** is a bonafide work of **"Shubh Dhar" (60003190057), Prajwal Bandewar" (60003190037), "Sabir Ali Baba" (60003190044)** and submitted in the partial fulfillment of the requirement for the award of the Bachelor of Technology in Information Technology.

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Dr. Vinaya Sawant
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Principal

Place: Mumbai

Date:

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Prof. Arjun Jaiswal
Name of the Guide

Name of the External Guide

Dr. Vinaya Sawant
Head of the Department

Dr. Hari Vasudevan
Principal

Place: Mumbai

Date:

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the Institute and can also evoke penal action from the sources, which have thus not been properly cited or from whom proper permission has not been taken, when needed.

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APPROVAL SHEET

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Shubh Dhar
Prajwal Bandewar
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Abstract

Machine learning is an emerging research field in crop yield analysis. Yield prediction is a very important issue in agriculture. Any farmer is interested in knowing how much yield he is about to expect. In the past, yield prediction was performed by considering farmer's experience on particular field and crop. The yield prediction is a major issue that remains to be solved based on available data. The predominant targets of this venture are expecting crop-yield which may be extraordinarily beneficial to farmers in making plans for harvest and sale of grain harvest. We plan to implement a system mastering set of rules that offers better prediction of appropriate crop for the corresponding vicinity and crop season in our country. The intention of this is the prediction of crops with a purpose to provide excessive yield inside the given area thinking about the weather and soil parameters. This is achieved by applying machine learning algorithms like Support Vector Machine and Decision Tree algorithm on agriculture data and recommends fertilizer suitable for every particular crop. Furthermore, the gathered data will then be conveniently stored and displayed on Cloud where it is easily accessible and modifiable.

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Plagiarism Report

1. Introduction

1.1 Motivation/Objective

India is an agricultural nation with a population of more than 2 billion people, of which about 70% of the population depends on agriculture. Agriculture is a major source of income for Indian citizens and agriculture has also had an impact on the Indian economy. Farmers have an immeasurable variety to choose from. fair products from soil cultures. Plant selection by the farmer The proposed system is a smart farming system that is automated to help farmers increase crop production by predicting which crops to grow. The system recognizes the soil quality and provides a list of selected general plants in a database. The system used various sensors that measured pH, soil moisture, temperature, and humidity. After the system for pH, soil moisture, humidity and temperature sensors has been designed, the next step is to create a model for the plant recommendation. Then the weather forecast is created. The Google API is used for the forecast to record the weather values in real time. Then, when checking weather conditions, recommend sowing. Due to the changing environmental conditions, the plant constantly shows the various diseases in order to control that the system recommends the solution to annihilate the disease. Farmers use the pesticide, fertilizer in the high The amount increased due to this toxicity of the soil to control this system of activity will recommend the best pesticide and fertilizer with the best measures.

The profit of agriculture does not always depend on normal wealth, but also on the information that is made available to the framework. , Type of fertilizer, climate and crops. These data sources will increase or express the profitability of any crop. The desire for habitual growing is not always specific and does not require subtleties of study. Information mining techniques and techniques are used to overcome these problems. machine learning. In the Department of Agriculture, information gathering is available for an awfully long period of time. must be searched in order to look at the data set and pay attention to the data it contains. The machine learning calculations are used to adopt the unique capacity of cultivation with the aim of activating fewer problems in the information sources and making profits. India is one of the main producers of agricultural products, but it has much less land. Productivity needs to be increased so that farmers get a higher wage from an equivalent parcel with much less work using specific and correct general observations like urine, fertilizer, soil, etc. at the right time for the belly to increase your productivity and increase your yields.

1.2 Major Challenges

There have been several major challenges and constraints that we had to face while developing our project model. They are:

i. Constantly changing weather conditions: Due to extreme shifts of weather between cold, hot and rains our sensors such as soil moisture sensor and ph sensor get effected and give redundant or inaccurate values.

ii. To train the farm and agricultural workers to use the website and implement the IOT system: Some of these users may not be educated enough to operate our system smoothly. Thus by giving them adequate training about what are sensors and gadgets are, along with which ones to use and how to record the values of the soil sample from the Arduino Uno microcomputer along with receiving their output on the website, we can resolve this issue.

iii. To train the farmers to use our interface: The farmers need to be educated about how to use our user-friendly website. The features of Weather Forecast, current crop market trends, crop and fertilizer prediction, etc can be made clear to them.

iv. Getting the adequate soil samples: Availability of soil samples other than the red soil is generally very low. Thus, our input samples have been fairly limited.

1.3 Report Overview

In this report, the Chapter 1 describes the motivation to pick up this problem statement and choose our solution which involves the issues faced by farmers regarding crop production and fertilizer usage.

Chapter 2 includes the analysis of present work based on our project where we detail their benefits and disadvantages. Inferences are made to judge whether the system worked or not.

Chapter 3 is where we gave a detailed overview of the Problem Definition and scope of the project. This is followed by explaining our proposed methodology and features and its advantages in today's market.

Chapter 4 involves the Project Management where we specify the timeline along with the breakdown of modules per month. We analyse the risks associated with developing our IOT system, while analysing the various feasibility studies. The various software and hardware project components are mentioned too along with the project estimation.

In Chapter 5, we analyse the System Design. Various illustrations have been displayed, like the Data Flow Diagrams, UML Diagrams as well as the System Architecture. The two major datasets of the crop and the fertilizer that have been used are described as well.

In Chapter 6, we move on to the implementation section where we give a brief working of the system and explain the algorithms that we have incorporated. Screenshots of various output screens in our website along with the live pictures of our system circuit and its setup have also been uploaded.

In Chapter 7, which is the last and final chapter, we give a detailed overview of our Testing schemes. Various test cases have been implemented and testing strategies used have been elaborated on as well. The Conclusion and the Future scope which will be implemented as an add-on have also been mentioned.

2. Literature Review

2.1 Existing Work

2.1.1 Literature Related To Existing Systems

Title	Authors	Advantages	Disadvantages	Result
Demand Based Crop Recommender System For Farmers	S. Kanaga Suba Raja, R. Rishi, E. Sundaresan, and V. Srijith	It increases the crop rate by 3%.	More sensors can be used in order to have a complete summary of the crop health, produce and the soil conditions as well. Battery energy consumption can be more as for every node, the sensors need the batteries.	By applying this technique, the farmers will be benefited and improvement in yielding of crops
RSF: A recommendation system for farmers	M. J. Mokarrama and M. S. Arefin	It increases the production rate by 7%	The use of an expert system and IoT is an excellent way to improve the crop rate.	In this paper based on the algorithm and data sets collected recommend ation of crops are made at upazila level.

Crop recommendation system for precision agriculture	S. Pudumalar, E. Ramanujam, R. H. Rajashree, C. Kavya, T. Kiruthika, and J. Nisha,	The use of an expert system and IoT is an excellent way to improve the crop rate.	The expert system does tell about the herbicide and pesticide recommendation depending on the disease but the use of a camera or an image capturing device is used in their paper. Also, battery consumption can be high.	In this technique, it was more productive and more profit and it helped many to plant the right crop at the right time.
Design of fertilization recommendation knowledge base and application	Z. Ren and X. Lu	It improves the precision.	The ubimote sensor board used can result in failures and this, in turn, may stop the entire system from working as this is the one which gathers and sends the information to the farmers.	From user similarity and service resource value the following theory was validated

Affordable field environmental monitoring and plant growth measurement system for smart agriculture	T. Okayasu et al	It increases IOT network accuracy.	The main disadvantage is that in case of large fields, the playhouses cannot be used as it increases the cost. The drip irrigation setup cost may also increase when we consider a large area.	The inclusion of trust into the RS increases both accuracy and coverage.

2.1.2 Literature Related to Methodology/Approaches/Algorithms

There are numerous literature to be had related to methodology and algorithms for soil and fertilizer predictor device. some of them are:

1) "Development of an expert machine for soil fertility recommendation the usage of fuzzy logic" by using Thayalini et al. (2015) - This paper affords an expert system that uses fuzzy logic to recommend soil fertility control practices.
2) "Prediction of soil nutrient reputation using machine learning strategies: A review" through Behera et al. (2021) - This review paper discusses the usage of ML algorithms and strategies for predicting soil nutrient condition.
3) "Development of a decision support system for fertilizer recommendation the usage of fuzzy logic" with the aid of Kumar et al. (2014) - This paper presents a decision support system that uses fuzzy logic to recommend fertilizer utility rates.
4) "Soil fertility prediction using machine learning algorithms" via Acharya et al. (2020) - This paper gives a observe that uses device studying algorithms to predict soil fertility.

5) "an internet-primarily based professional device for soil fertility advice the usage of artificial neural network" by means of Zaman et al. (2012) - This paper provides an internet-based professional gadget that uses artificial neural networks to suggest soil fertility management practices.

2.1.3 Literature Related to Technology/Tools/Frameworks

There are various literature to be had associated with era and frameworks for soil and fertilizer predictor gadget. some of them are:

1) "Sensor-based intelligent agricultural decision support system for crop control" through Singh et al. (2019) - This paper affords a sensor primarily based wise agricultural choice guide system for crop management that contains soil and fertilizer control.
2) "Internet of factors based total Smart Agriculture gadget: Framework, Implementation and demanding situations" by way of Thakur and Sharma (2018) - This paper offers a framework for an Internet of Things (IoT) based smart agriculture system that includes soil and fertilizer management.
3) "Farm decision assist systems: An overview of literature" by Läpple et al. (2015) - This assessment paper discusses numerous farm decision support systems, consisting of the ones associated with soil and fertilizer management.
4) "Agriculture information system: a software for Big Data in agriculture" through El-Hassany et al. (2018) - This paper discusses the use of big data and statistics systems in agriculture, including for soil and fertilizer management.
5) "Agriculture 4.0: The future of Farming generation" by means of Thakur and Sharma (2019) - This paper gives an overview of Agriculture 4.0, consisting of the usage of technology for soil and fertilizer management.

Chapter 3

Proposed Methodology/Approach

3.1 Problem Definition

To design an Internet of Things(IOT) based farmer-friendly model which can gauge various soil traits like moisture, nutrients, temperature, etc and determine the crop yield which can help decide whether that particular soil is suitable to grow the crop or not.

Machine learning is an emerging research field in crop yield analysis. Yield prediction is a very important issue in agriculture. Any farmer is interested in knowing how much yield he is about to expect. In the past, yield prediction was performed by considering farmer's experience on particular field and crop. The yield prediction is a major issue that remains to be solved based on available data. The predominant targets of this venture are expecting crop-yield which may be extraordinarily beneficial to farmers in making plans for harvest and sale of grain harvest. We plan to implement a system mastering set of rules that offers better prediction of appropriate crop for the corresponding vicinity and crop season in our country. The intention of this is the prediction of crops with a purpose to provide excessive yield inside the given area thinking about the weather and soil parameters. This is achieved by applying machine learning algorithms like Support Vector Machine and Decision Tree algorithm on agriculture data and recommends fertilizer suitable for every particular crop.

3.2 Scope

The machine makes use of supervised and unsupervised Machine learning algorithms and offers first-rate end result primarily based totally on accuracy. It will act as a medium to deliver the farmers green records required to induce high yield and as a consequence maximize income which successively will reduce depression amongst farm workers(which is a prevalent issue faced by India) and reduce their problems.

3.2.1 Existing System

In maximum of the existing methods, the process of finding the soil type, identifying the leaf disease and who prefer the fertilizer had been all carried out manually. The technique became susceptible to various disadvantages. Even when the framework became digitalized, it has certain issues as, predicting a numerous fertilizer for a soil kind, sure files concerning the leaf sickness or soil type or fertilizer won't be updated.

3.2.2 Disadvantages of Existing System

1. System is not précised.
2. Detecting area is limited.

3.3 Proposed Approach to build

Crop and weather prediction:

Precision agriculture facilitates in reduction of non-suitable crop that so will growth productivity, apart from the following benefits like efficacy in enter in addition as output and better selection making for farming. This approach offers answers like providing an advice machine through an ensemble model with majority preference strategies mistreatment random tree, SVM and Naive Bayes as learner to recommend suitable crop supported soil parameters with excessive precise accuracy and potency. The categorized image generated via way of means of those strategies includes floor reality statistical data and parameters of it vicinity unit weather, crop yield, state and district sensible crops to expect the yield of precise crop under particular weather condition. The layout consists of collection period of time records and constructing the prediction model so creating a user interface for giving inputs. On the begin data preprocessing is completed. Once the pre-processing is completed, the usage of ML. set of rules model is generated for prediction. The test records is given to the generated model for prediction. The model is examined towards random enter values supported accuracy rate and error created while predicting throughout testing. Until the error price is decreased and accuracy rate is improved this approach is repeated. In order to accumulate the input from user, web application is employed. The inputs accrued are given to the crop prediction model for predicting the appropriate crops. The data set used for the weather forecast is collected by the NRSA Hyderabad station, which consists of five years of data on

meteorological parameters such as temperature, humidity, hours of sunshine, wind speed, wind direction, etc. In order to train the prediction model, data preprocessing is first carried out, which includes filling in missing values, data transformation and data normalization. Missing data values are filled in using the linear interpolation method. The hourly data is then converted into daily average data consisting of the minimum temperature and the maximum temperature calculated from temperature data. The Min max normalization is carried out on the data set in order to convert it into a common scale. For the time prediction, neural network models (NN) are used, which are a variant of the neural network (NN) that is the dependence of RNN over conventional NN, the Output from the previous step is forwarded to the current step, so the hidden state in RNN remembers the order, while in the conventional neural network all values are handled independently of yes. RNN consists of an input layer, two hidden layers and an output layer.

Crop disease and fertilizer prediction:

A digital camera or similar device is used to take different types of pictures and then they are used to identify the affected area on the leaves, then different types of image processing techniques are applied to them, processing these pictures to obtain different ones and useful functions required for this purpose. The identification of plant leaf diseases is particularly necessary in order to predict both the quality and the quantity of the first segmentation step, which is mainly based on a smooth polygonal leaf model that is first created and then used to calculate the development of an energetic contour By combining the global shape descriptors of the polygonal model with local curvature features, the leaves on the back are classified in data sets. The first segmentation step based on the graphical cutting approach is carried out first and then used for guidance. The development of leaf boundaries and implementation of classification algorithms to classify diseases and recommend fertilizers for affected leaves. The leaves are attacked by bacteria, fungi, viruses and other insects. The algorithm classifies the image of the sheet as normal or affected. like color, shape, textures. Then hyperplane was created with conditions to categorize the preprocessed leaves and also to implement a multiclass classifier to predict diseases in the leaf image with greater precision. Recommend the fertilizer for the affected leaves depending on the severity. Fertilizers can be organic or inorganic. The manager can store the

fertilizers according to the disease classification with severity levels. Recommended fertilizer measurements based on the severity of the disease.

Market Analysis and policies recommendation:

The dataset is prepared by collecting crop and rainfall data. There are many datasets that contain data. We got the data containing the details of the whole price index and monthly rainfall of the different crops. One of the most important steps in any machine learning project is data cleansing. There are several methods of statistical analysis and techniques for visualizing the data in the dataset that you can use to explore the data to identify the appropriate data cleansing operations to perform. There are some very simple data cleansing operations before we move on to the advanced methods that we can carry out in a machine learning project on each dataset. They are so important that models can break or report overly optimistic success results if they are missing. In our dataset, we cleaned up all null values and checked if all data types are valid. Also called EA, exploratory data analysis is a very important step in finding and investigating various datasets and summarizing their important characteristics, often using different data visualization methods.and conclusions to decide how best to monitor data sources to obtain results with greater precision. We have tested about 6 different algorithms which are the most efficient. From the dataset, select all rows and columns from 1,2,3 to "X" which is the independent variable. From the dataset, select all rows and column 4 to "y" What is the dependent variable. Variables x and y with a decision tree regressor Updates the user interface with the predicted values The predicted WPI data is converted to price per hundredweight using the formula (WPI x base price) / 100 and displayed visually in an understandable web application created using the Flask framework. The bottle is a popular extensible web micro framework for building beautiful web applications with Python. An index page is created and from there we can navigate to 20 different cultures and see their predictions in detail. Then according to market condition we recommend policies to farmer which useful for them.

Feature description:

The key function for plant increase is nutrients. The soil includes differing types of nutrients. Soil enables in maintaining water. This could facilitate in accurate nutrient provide to the plant increase. The usage of moisture detector, the function soil moisture is measured from diverse soil. This enables us in information the need of moisture content material for the real soil. Mistreatment temperature detector, the temperature of the soil is measured that suggests the fine temperature for plant increase. Moisture level in the air is measured with humidity that is measured mistreatment humidness detector. The readings measured from the sensors vicinity unit measured and used for processing. The ph of the soil is a totally vital function as it affects the plant increase in lots of methods that. The fine pH range for the soil is 5.5-7.0. The soil have to be in the prescribed ph range to attend to the plant increase.

The developed system allows in overcoming the drawbacks determined in the present system. The approaches in the proposed device consists of growing the yield of crops, duration evaluation of crops victimization ML, selecting efficient parameters, growing smarter choices and getting higher yield. Effective algorithms were given for use for early prediction of crops. By choosing accurate elements maximum beneficial records are regularly generated the usage of ml model with the aid of using intake of numerous parameters and this generated statistics allows the farmers with the aid of using suggesting a proper crop to be seeded. To endorse the correct crop for a particular field area is that the primary aim of crop recommending model. By choosing the appropriate crops for the field space we can limit the lack of crops. suitable algorithms with selective options needs to be selected for the reason that accuracy stage of suggesting crops would possibly range supported the type of components selected has be selected. Machine Learning is determined be the most powerful technology for predicting appropriate crops and its yield.

3.3.1 Benefits of Proposed Solution

1) Fertilizers or vitamins are encouraged for all of the crop suggestions, this could facilitate the farmers to induce the proper information to grow the plants and enhance the productiveness and income.

2) Our projected system works with minimal hardware implementation.

3) AI is supposed to provide unique and extra accurate end result with minimal delay.

4) This technique is probably a value efficient and without a doubt on hand decision for farmers.

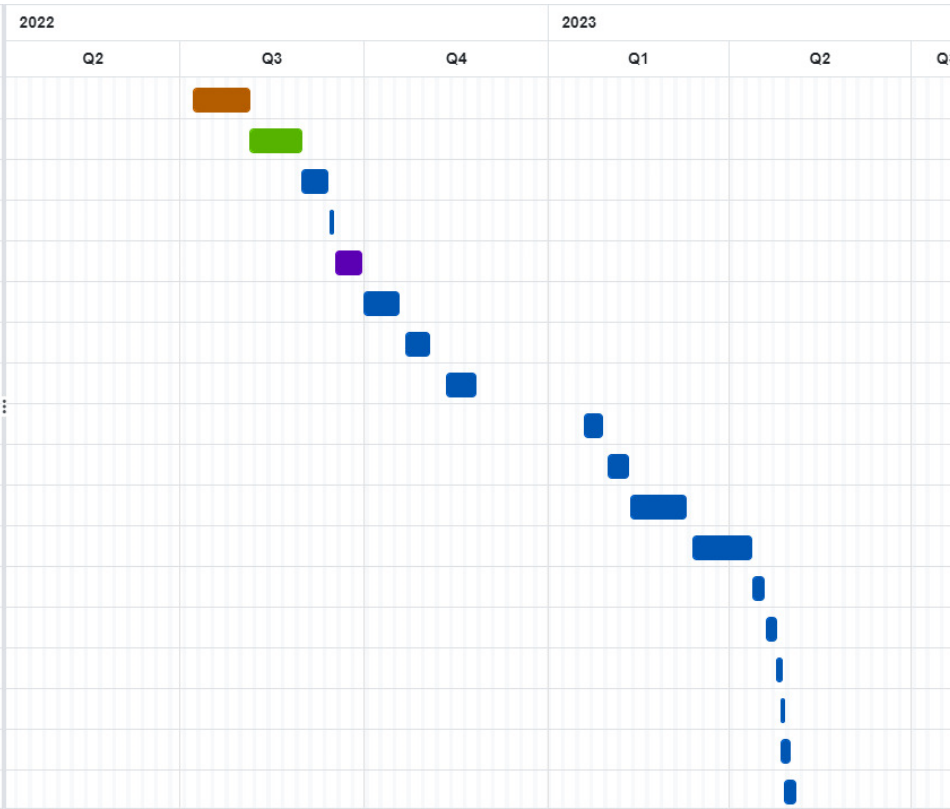
Chapter 4

Project Management

4.1 Project Schedule

4.1.1 Timeline Chart

ID	Name
1	Project Planning Phase
2	Assessment
3	Project Arrangement
4	Task Delegration
5	Project Design Phase
6	Wireframe
7	Mock up
8	Review
9	Project Development Phase
10	Framework
11	Coding
12	Project Testing Phase
13	Model Accuracy Testing
14	UI Testing
15	System Testing
16	Project Deployment Phase
17	Backend Hosting
18	Model Hosting



4.2 Feasibility Study

This system is possible for all health care department like science lab hospital and clinic etc and this method can use while not specialists in this field anyone can use who have data concerning using online services which is able to facilitate to use this method any generation folks can use this method on the laptop.

4.2.1 Technical Feasibility

The framework ought to be assessed from the specialized reason for read first the evaluation of this practicability ought to be upheld a rundown kind of the framework interest inside the provisions of info yield projects and techniques having known an outline framework the examination ought to keep up to suggest the kind of pack required approach building up the framework of running the framework whenever it has been planned

- Technical issues raised during the investigation are:
- Is the existing technology sufficient for the suggested one?
- Can the system expand if developed?

the undertaking should be created indicated the predetermined capacities and execution are accomplished among the limitations the task is created among most recent innovation through the innovation may become old once some measure of some time due to the specific undeniable truth that never form of same code upholds more seasoned variants the framework should in any case be utilized hence there are marginal imperatives included this task the framework has been created exploitation java the undertaking is in fact feasible for advancement

4.2.2 Economic Feasibility

The creating framework ought to be even by worth and benefit. Measures to confirm that exertion is focused on a project, which may give best, come at the most punctual. one through and through the variables, that affect the occasion of a new framework, is that the value it'd need. The following are assortment of the necessary cash questions asked all through starter examination:

- The costs conduct a full system investigation.
- The cost of the hardware and software.
- The benefits in the form of reduced costs or fewer costly errors

Since the framework is created as a neighborhood of task work, there is no manual worth to purchase the projected framework. Furthermore every one of the assets are as of now available, it offers an image of the framework is financially feasible for improvement.

.

4.3 Project Resources

4.3.1 Hardware Requirements

1. Processor: Intel Core i3 or more.

2. RAM: 4GB or more.

3. Hard disk: 250 GB or more.

Sensors used for analysis:

4. Soil moisture Sensor
5. Mq2 Gas Sensor
6. LDR Sensor
7. Raspberry pi
8. Arduino Uno
9. pH sensor

4.3.2 Software Requirements

1. Operating System : Windows 10, 7, 8.
2. Python.
3. Anaconda.
4. Spyder, Jupyter notebook, Flask.
5. MYSQL.

4.4 Technologies Used

Python

Python is an interpreter, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed. This is the object-oriented programming language that we have used to implement the machine learning models and make predictions. It is also used to measure their precision and accuracy.

Anaconda

It is a free and open source distribution of the Python and R programming languages for data science and machine learning related applications (large-scale data processing, predictive analytics, scientific computing), that aims to simplify package management and deployment. Package versions are managed by the package management system conda. The Anaconda distribution is used by over 6 million users, and it includes more than 250 popular data science packages suitable for Windows, Linux, and MacOS.

MySQL/SQLYog

MySQL is well known as world's most widely used open-source database (back-end). It is most supportive database for PHP as PHP-MySQL is most frequently used open-source scripting database pair. The user-interface which WAMP, LAMP and XAMPP servers provide for MySQL is easiest and reduces our work to a large extent. This is the open-source database we have used for the backend. A track/log of registered users is maintained in the form of tables.

Flask

We have used Flask and it's libraries to create web pages for this project. It is mainly used for the backend functionalities of pages like the user login/registration page, services page, etc.

Pickle File

We have used Flask and it's libraries to create web pages for this project. It is mainly used for the backend functionalities of pages like the user login/registration page, services page, etc.

TensorFlow

This is an important machine learning platform we have used to import Keras. Keras is used to implement neural networks for crop and fertilizer prediction efficiently.

HTML/CSS

We have used HTML and CSS for the frontend to make the web pages and also style and layout them using appropriate font, color, spacing, etc.

JavaScript

This is another front-end scripting language that we have used to make the web pages more dynamic and interactive, control multimedia, animate images, etc.

4.5 Project Estimation

Software Module	LOC
Training	250
Command Interface	1200
Model Training	250
Model Testing	1000
Graphical User Interface	500
Setting	800
Total Estimated Lines of **Code (LOC)**	**4000/Person**

COCOMO Estimation Model

The Basic COCOMO formula takes the form:

Effort, E = a(KLoC/KDSI)^b person months

Duration, D = c(E)^d months

Person, P = E/D persons

where **E** is the effort applied in person-months, **KLoC** is the estimated number of thousands of delivered lines of code for the project, **D** is total time duration to develop the system in months, and **P** is number of persons required to develop that system.

Software Module	LOC
Graphical User Interface	500
Command Management	500
OS Interface	500

Model Training	500
Test Module	250
Integrating Module	250
Integrating Backend and Frontend	1000
Action Commands Specification	500
Total Estimated Lines of Code (LOC)	4000

Total lines of code for the proposed system will be approximately **4000/ Person**.

Effort Estimation

The system falls into the **embedded** category. The value a and b according to embedded system is a = 2.8 and b = 1.20

Total LOC (approx.) of project is 5000LOC = 5.00KLOC

Effort (E) = $a(\text{KLoC})^b$

$E = 3.6*(5.00)^{1.20}$

E = 24.83 ≈ 25 Person Months

Duration Estimation

The value c and d according to embedded system is c = 2.5 and d = 0.32

Duration (D) = $c\ (E\ )^d$

$D = 2.5*(24.83)^{0.32}$

D = 2.5 *2.7950

D = 6.98 ≈7 Months

Person Required

Person Required = Effort Applied (E) / Development Time (D)

= 24.12/7

= 3.44 ≈ 3 Persons

Estimated Cost of System

We assume each team member charges ₹500/- per month, ₹1000/- required for other resources & miscellaneous purposes, and ₹500/- for Microphone. Thus,

Estimated Cost of System = ((Person Charges * Person Required) + Resource Charges) * Duration [+ Hardware Cost]

= ((500 * 3) + 1000) * 7 + 500 = **₹18,000/-**

Estimation Summary

Summary of different estimation.

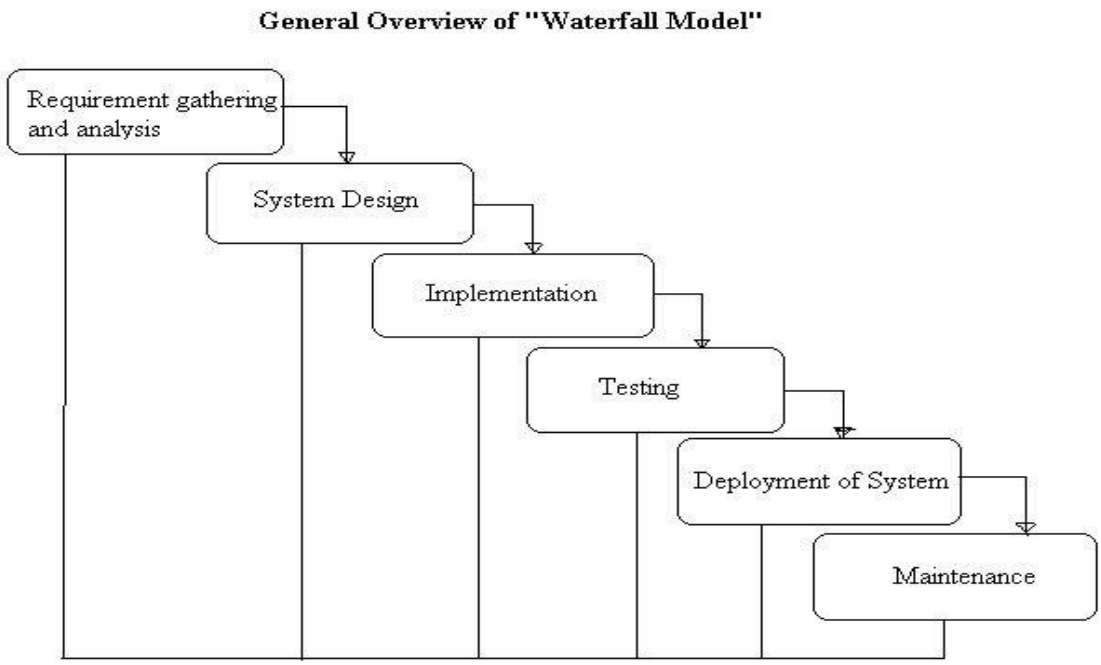
Estimation	Value
Size of the Project	7000 LoC
Effort Required	1 Person Months
Duration Required	12 Months
Person Required	3
Cost Required	₹12,000/-

Chapter 5

System Design

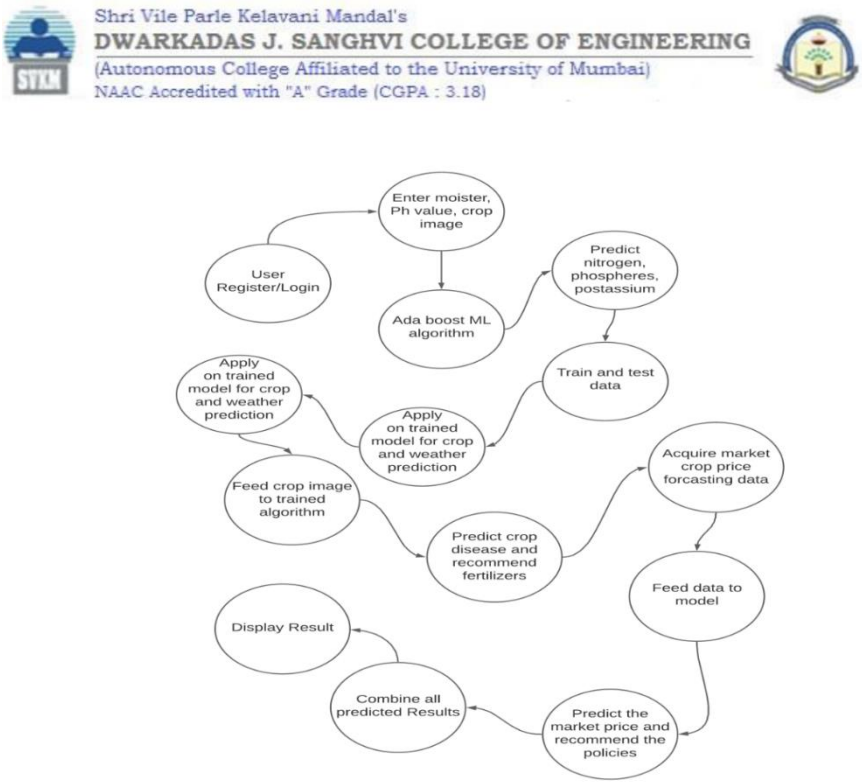
5.1 Design Diagrams

5.1.1 Waterfall Model

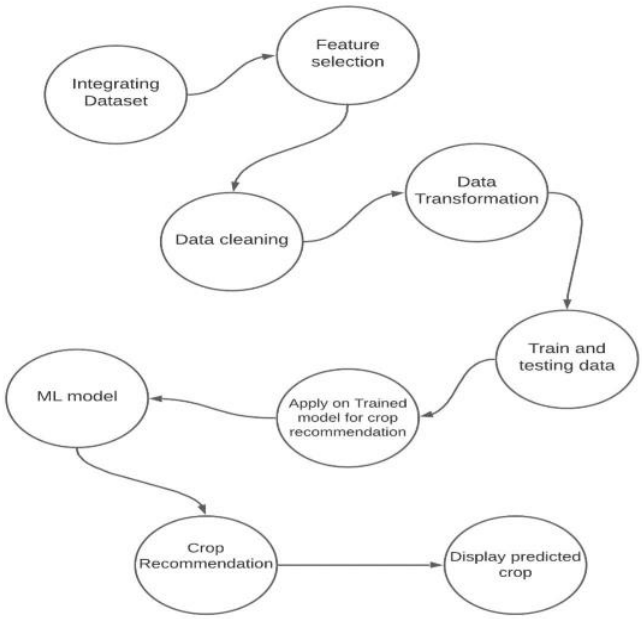


5.1.2 DFD Diagrams

DFD Level 1

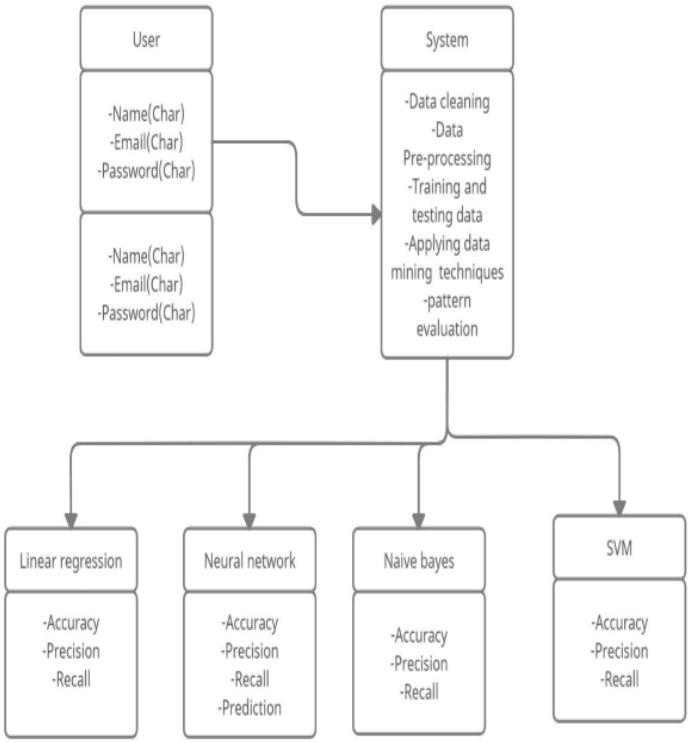


DFD Level 2

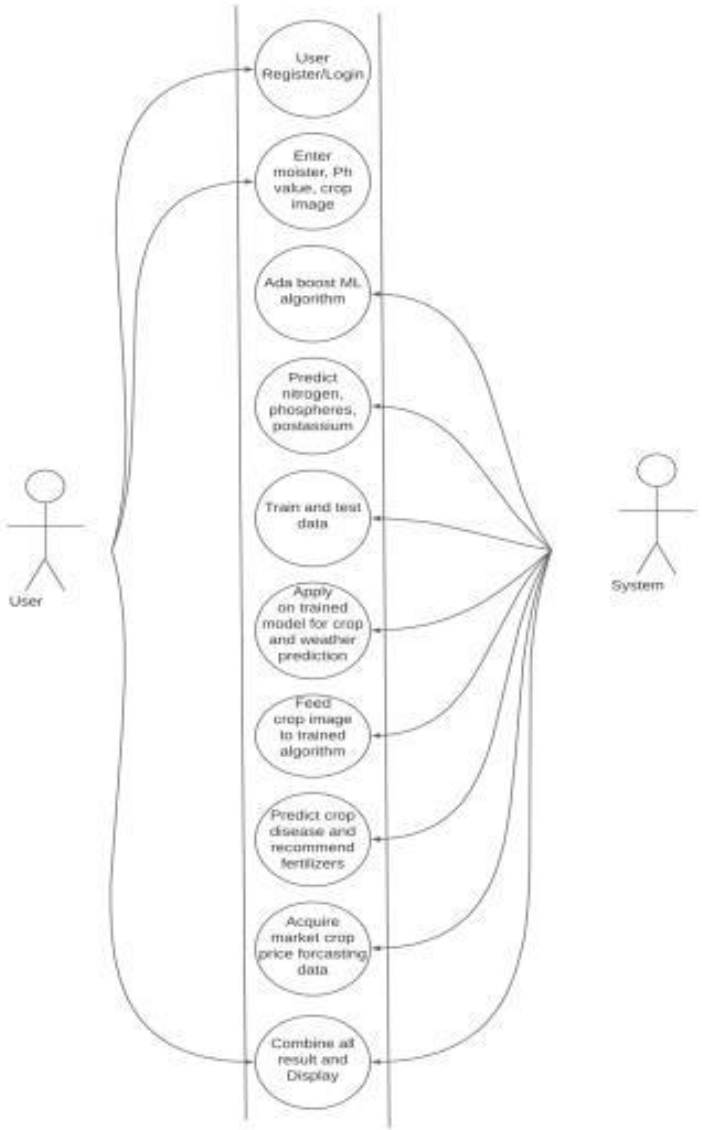


5.1.3 UML Diagrams:

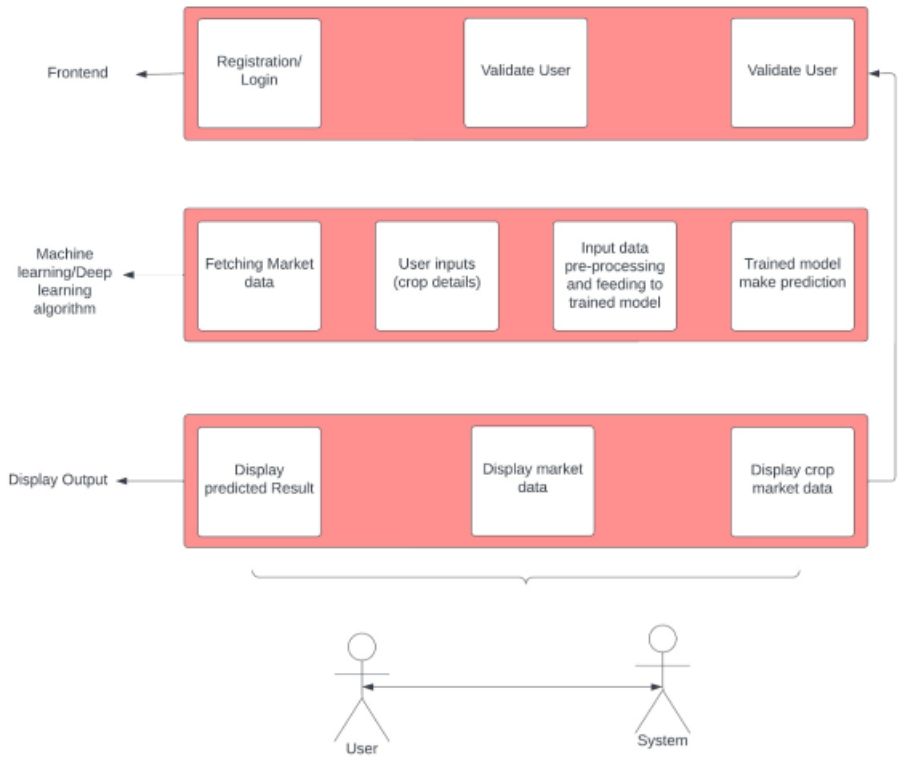
Class Diagram

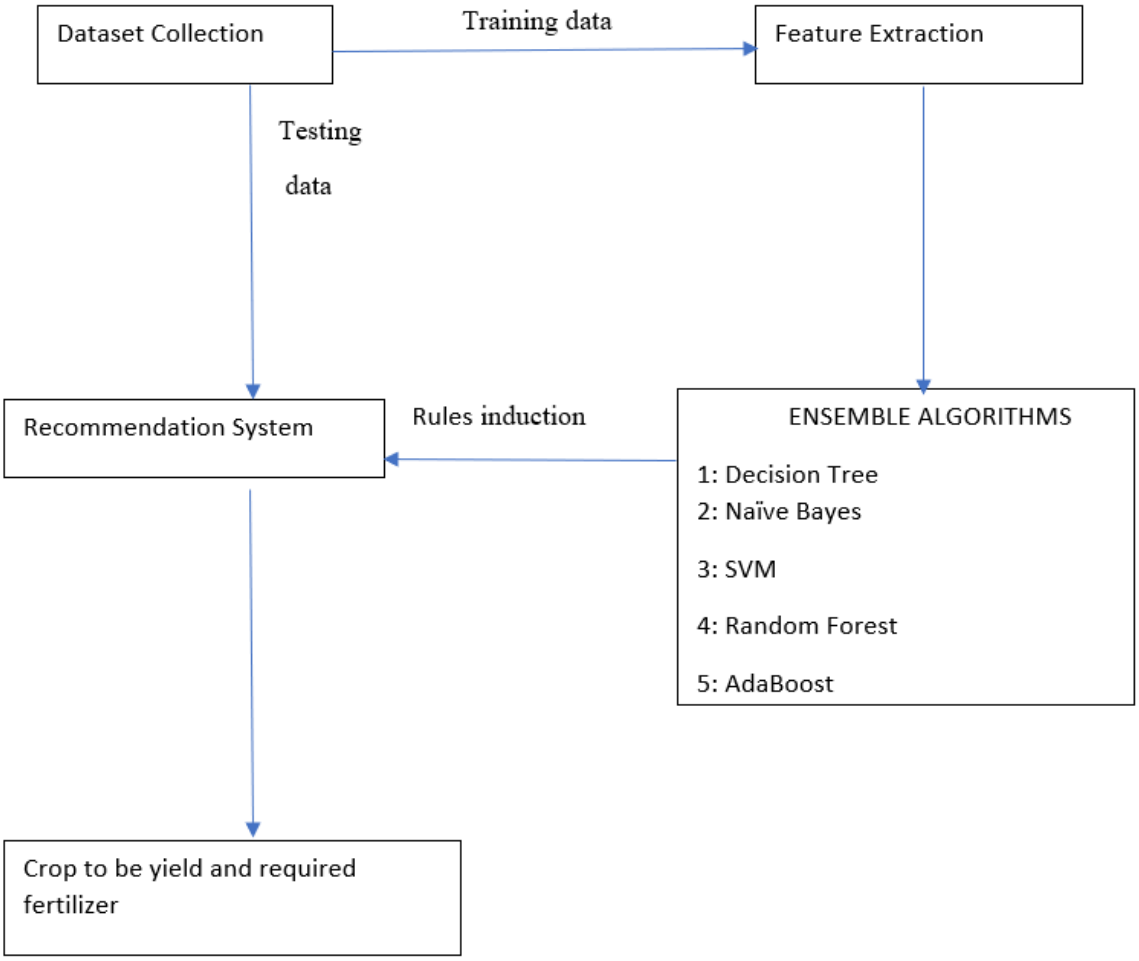


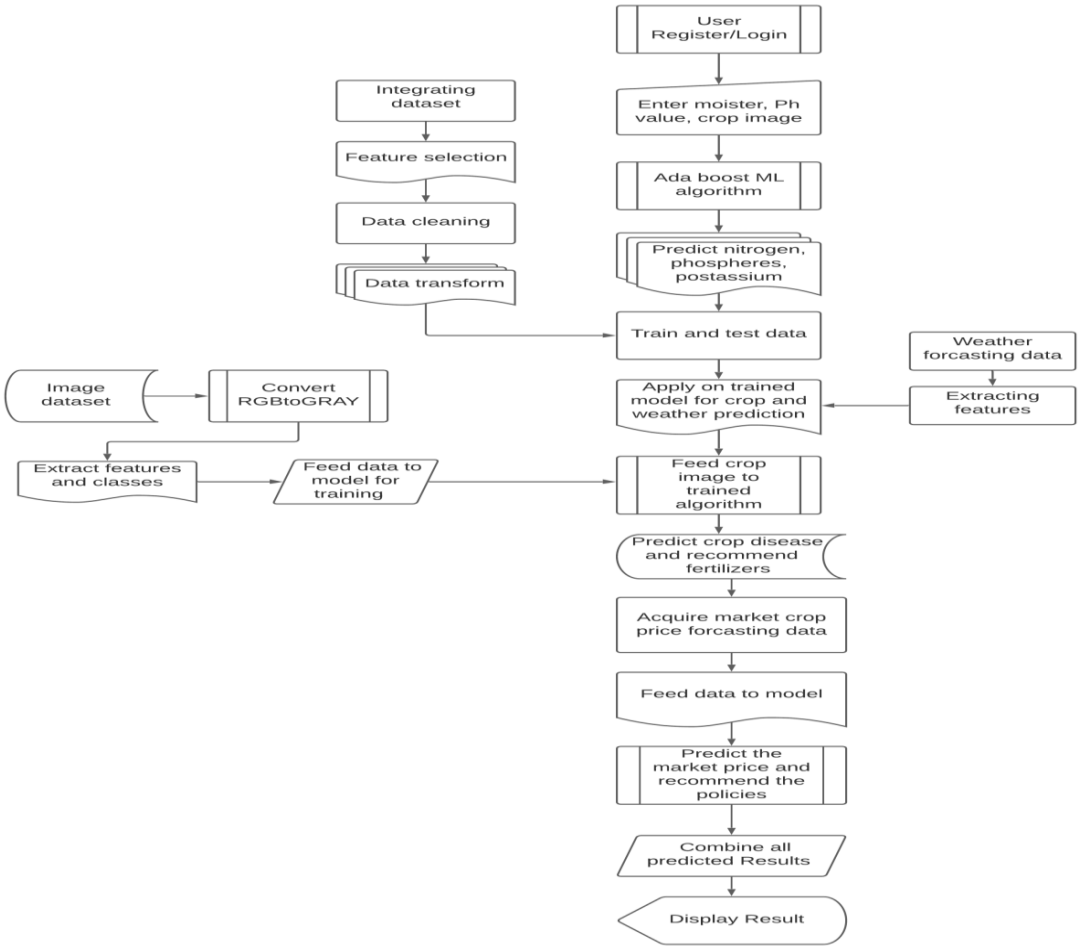
Use case diagram:



5.2 Proposed System Architecture




Dataset Collection
Training data
Feature Extraction
Testing
data
Recommendation System
Rules induction
ENSEMBLE ALGORITHMS
1: Decision Tree
2: Naïve Bayes
3: SVM
4: Random Forest
5: AdaBoost
Crop to be yield and required fertilizer



5.3 Description of Datasets

Datasets used

1) **Crop recommendation dataset**: This is the crops dataset we have used based on which we have trained our model for making further predictions based on n,p,k values. It consists of 2200 records. https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset

	A	B	C	D	E	F	G	H
1	N	P	K	temperatu	humidity	ph	rainfall	label
2	90	42	43	20.87974	82.00274	6.502985	202.9355	rice
3	85	58	41	21.77046	80.31964	7.038096	226.6555	rice
4	60	55	44	23.00446	82.32076	7.840207	263.9642	rice
5	74	35	40	26.4911	80.15836	6.980401	242.864	rice
6	78	42	42	20.13017	81.60487	7.628473	262.7173	rice
7	69	37	42	23.05805	83.37012	7.073454	251.055	rice
8	69	55	38	22.70884	82.63941	5.700806	271.3249	rice
9	94	53	40	20.27774	82.89409	5.718627	241.9742	rice
10	89	54	38	24.51588	83.53522	6.685346	230.4462	rice
11	68	58	38	23.22397	83.03323	6.336254	221.2092	rice
12	91	53	40	26.52724	81.41754	5.386168	264.6149	rice
13	90	46	42	23.97898	81.45062	7.502834	250.0832	rice
14	78	58	44	26.8008	80.88685	5.108682	284.4365	rice
15	93	56	36	24.01498	82.05687	6.984354	185.2773	rice
16	94	50	37	25.66585	80.66385	6.94802	209.587	rice
17	60	48	39	24.28209	80.30026	7.042299	231.0863	rice
18	85	38	41	21.58712	82.78837	6.249051	276.6552	rice
19	91	35	39	23.79392	80.41818	6.97086	206.2612	rice
20	77	38	36	21.86525	80.1923	5.953933	224.555	rice

2) **Fertilizer prediction dataset:** This is the dataset we have used for fertilizer prediction based on n,p,k values. It consists of 99 records.

https://www.kaggle.com/datasets/gdabhishek/fertilizer-prediction

	A	B	C	D	E	F	G	H	I
1	Temparatu	Humidity	Moisture	Soil Type	Crop Type	Nitrogen	Potassium	Phosphorc	Fertilizer Name
2	26	52	38	Sandy	Maize	37	0	0	Urea
3	29	52	45	Loamy	Sugarcane	12	0	36	DAP
4	34	65	62	Black	Cotton	7	9	30	14-35-14
5	32	62	34	Red	Tobacco	22	0	20	28-28
6	28	54	46	Clayey	Paddy	35	0	0	Urea
7	26	52	35	Sandy	Barley	12	10	13	17-17-17
8	25	50	64	Red	Cotton	9	0	10	20-20
9	33	64	50	Loamy	Wheat	41	0	0	Urea
10	30	60	42	Sandy	Millets	21	0	18	28-28
11	29	58	33	Black	Oil seeds	9	7	30	14-35-14
12	27	54	28	Clayey	Pulses	13	0	40	DAP
13	31	62	48	Sandy	Maize	14	15	12	17-17-17
14	25	50	65	Loamy	Cotton	36	0	0	Urea
15	32	62	41	Clayey	Paddy	24	0	22	28-28
16	26	52	31	Red	Ground Nu	14	0	41	DAP
17	31	62	49	Black	Sugarcane	10	13	14	17-17-17
18	33	64	34	Clayey	Pulses	38	0	0	Urea
19	25	50	39	Sandy	Barley	21	0	19	28-28
20	28	54	65	Black	Cotton	39	0	0	Urea

Chapter 6

Implementation

6.1 Working of System

Our system works as a crop classification model using Python and Anacondas and into categories labelled rice, maize, chickpea, kidneybeans, pigeonpeas, mothbeans, mungbean, blackgram, lentil, pomegranate, banana, mango, grapes, watermelon, muskmelon, apple, orange, papaya, coconut, cotton, jute and coffee.

6.2 Description of Algorithms to be used (Modular Description)

- Machine learning classification:-

Following algorithms will be applied on features obtained above.

Decision tree :-

Decision tree fabricate characterization or relapse models inside the plan of a tree structure. It separates a dataset into more modest and more modest subsets though at steady time a related Decision tree is gradually evolved. the final word result's a Decision tree with call hubs and leaf hubs. a decision hub (e.g., Outlook) has 2 or extra branches (e.g., Sunny, Overcast and Rainy). Leaf hub (e.g., Play) addresses an arrangement or call. The highest choice hub during a Decision tree that relates to the solitary indicator referenced as root hub. Decision trees will deal with each downright and mathematical information.

Naïve bayes: -

In system learning, Naive Bayes classifiers are a family of clean probabilistic classifiers supported by applying Bayes Theorem with sturdy (naive) independence assumptions between the alternatives. Naive bayes will be a smooth method for building classifiers: models that

assign elegance labels to disadvantage instances, described as vectors of feature values, anyplace the class labels are drawn from some finite set. It's not one algorithmic application for coaching such classifiers, however a circle of relatives of algorithms supported a popular principle: all naive bayes classifiers assume that the worth of a specific function is independent of the real worth of the alternative characteristic, given the class variable. Naive bayes classifiers are a preferred applied mathematics method of email filtering. They emerged within the middle of the 90s and had been one among the number one attempts to tackle unsolicited mail filtering downside. Naive bayes usually use a bag of phrase alternatives to identify spam email, a method usually hired in text category. Naive bayes classifiers paintings by way of correlating the utilization of tokens (generally phrases, or usually different constructions, grammar or now not), with spam and non-junk mail e-mails so exploitation bayes theorem to calculate a hazard that an email is or isn't always a junk mail message. We tend to use a terribly similar approach for fake information detection, and it'll be represented in extra info inside the next chapter.

SVM: -

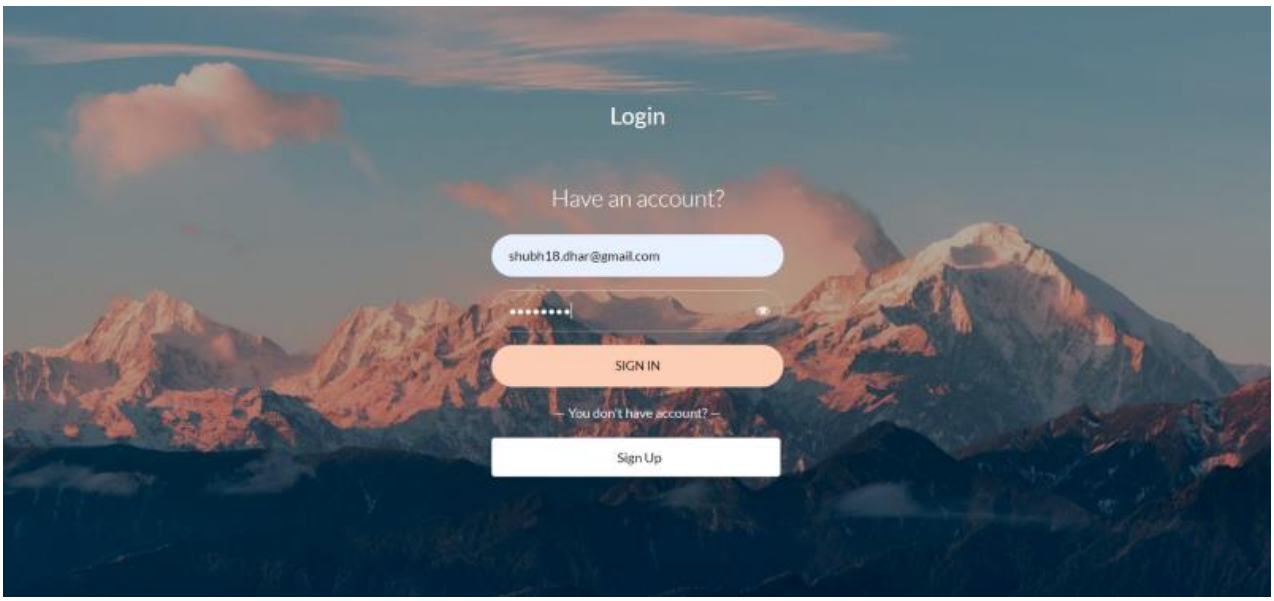
On this step, the set P and RN are wont to teach the biased svm classifier. As soon as faced with the problem of unbalanced dataset category, the most downside encountered by way of the svm technique is that the type surface is honestly on the brink of the few magnificence, mainly of incorrect type. Take dichotomies for example, if the educational pattern quantitative relation of fantastic class and negative category is 100:1, then the general classifier has a tendency to classify all of the samples as superb magnificence, therefore getting a higher type accuracy. But, this approach isn't always high-quality in employment. Inside the case of willing facts, the placing of the penalty difficulty will effectively reduce the effect of inclined data on the education approach. Here, the strategy in literature is employed to line the penalty trouble, and therefore the quantitative relation of the penalty problem of nice to negative instances is successful the quantitative relation of the pattern size of negative to nice cases severally.

Random forest: -

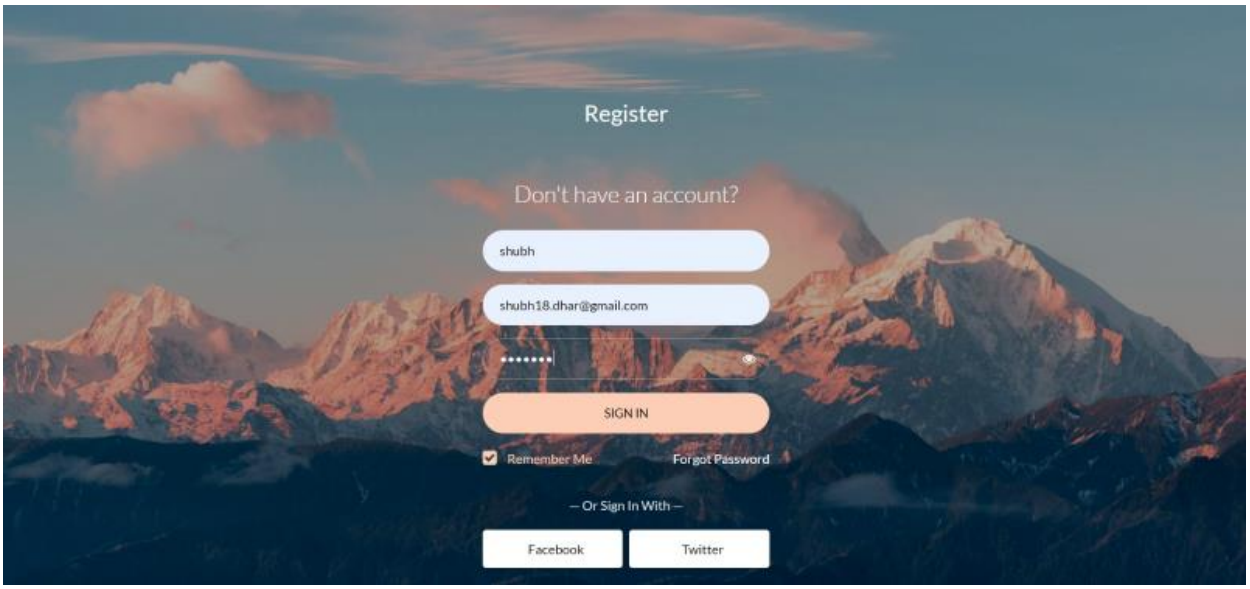
Grouped random forest(grf) methodology is used to identify intrusions inside the network through dividing the complete network into character systems in order that eager remarks may be executed at the communication initiated within the network. This method selects and monitors each node in the community randomly from any or a couple of places. This is the purpose, this proposed technique is known as group random forest (grf). Whilst a community is hooked up and communication is initiated, then data is transferred from node to another node from sender to vacation spot. This procedure is sustained till transfer of statistics is completed. Whilst conversation is going on, log files are created and then those log files are given as input for the proposed technique.

6.3 Interface Design

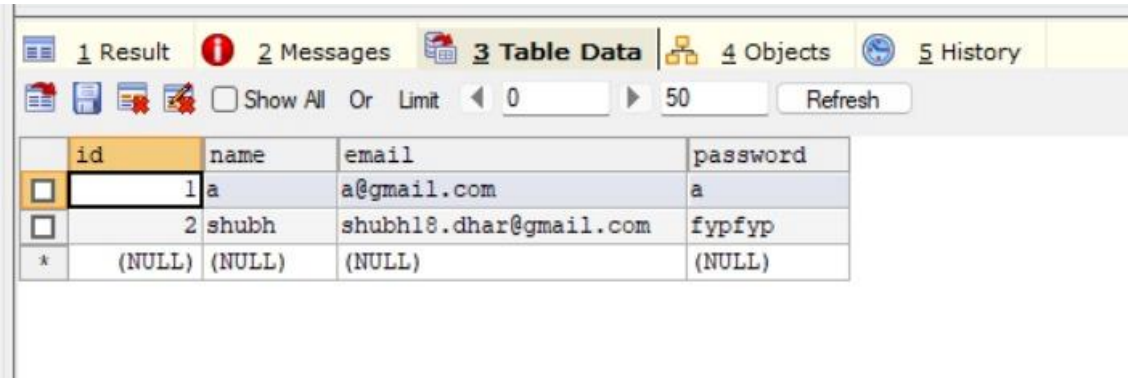
Website Login Page



Website Register Page

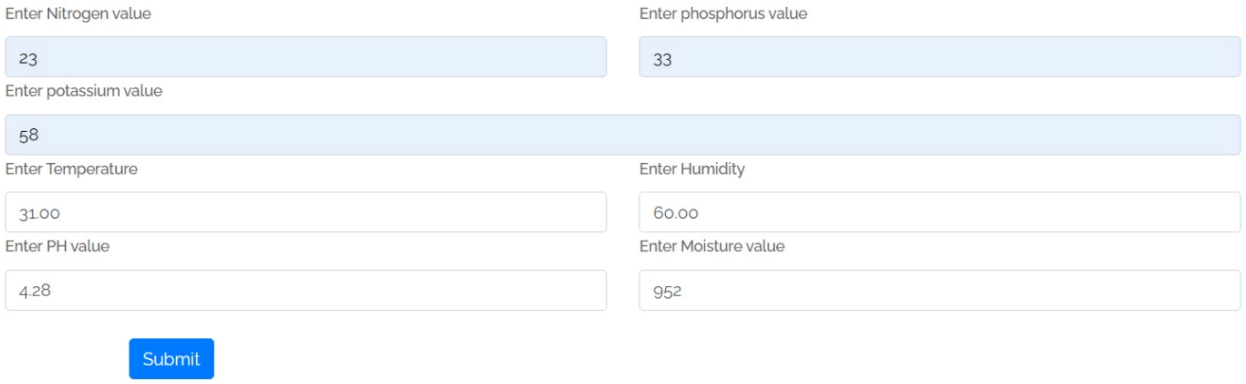


User Database Record



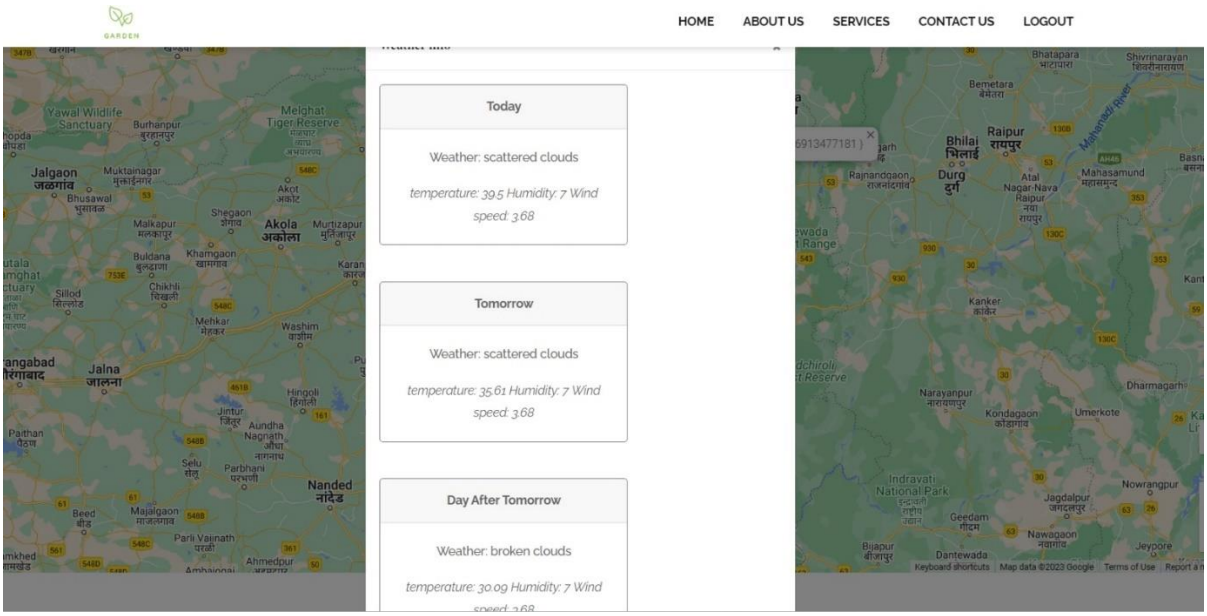
	id	name	email	password
	1	a	a@gmail.com	a
	2	shubh	shubh18.dhar@gmail.com	fypfyp
*	(NULL)	(NULL)	(NULL)	(NULL)

Services Prediction Page after entering values



Crop and Fertilizer Output

Weather Forecast Page



Market Analysis Page

Comodity	Variety	Maximum Price	Average Price	Minimum Price	Last Updated On
Apple	Simla	₹ 20,000	₹ 12,000	₹ 8,000	03 May 2023
Chikoos(Sapota)	Other	₹ 1,500	₹ 1,100	₹ 1,000	03 May 2023
Grapes	Black	₹ 5,000	₹ 3,750	₹ 2,500	03 May 2023
Grapes	White	₹ 3,100	₹ 2,500	₹ 1,850	03 May 2023
Guava	Guava	₹ 4,000	₹ 2,000	₹ 1,500	03 May 2023
Karbuja(Musk Melon)	Karbhuja	₹ 1,400	₹ 1,200	₹ 800	03 May 2023
Mango	Badami	₹ 5,550	₹ 2,750	₹ 800	03 May 2023
Mango	Dusheri	₹ 11,110	₹ 4,444	₹ 2,222	03 May 2023
Mango	Totapuri	₹ 3,611	₹ 1,667	₹ 833	03 May 2023
Mousambi(Sweet Lime)	Mousambi	₹ 4,520	₹ 3,300	₹ 1,210	03 May 2023

Farming Trends Page

PM-Kisan Scheme

Pradhan Mantri Kisan Samman Nidhi Yojana is an initiative of the Government wherein 120 million small and marginal farmers of India with less than two hectares of landholding will get up to Rs. 6,000 per year as minimum income support. PM-Kisan scheme has become operational since 1st December 2018. Under this scheme, cultivators will get Rs. 6000 in three installments.

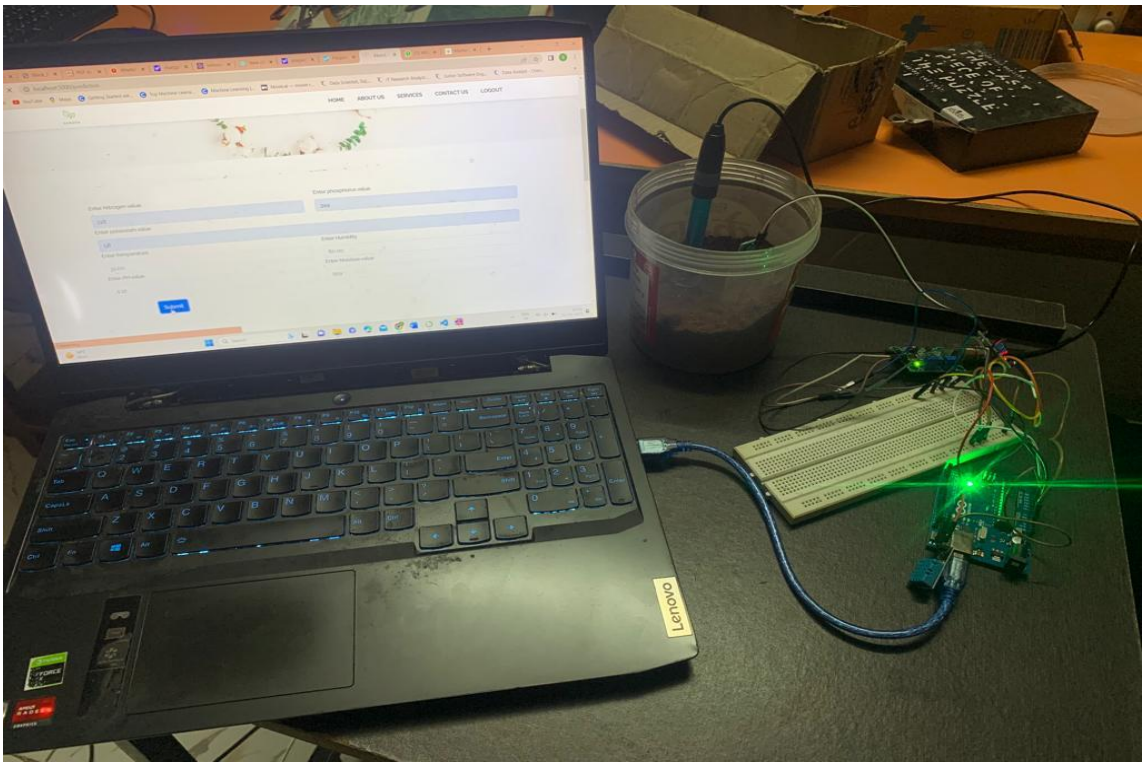
For more details Click Here!

Pradhan Mantri Kisan Maandhan yojana

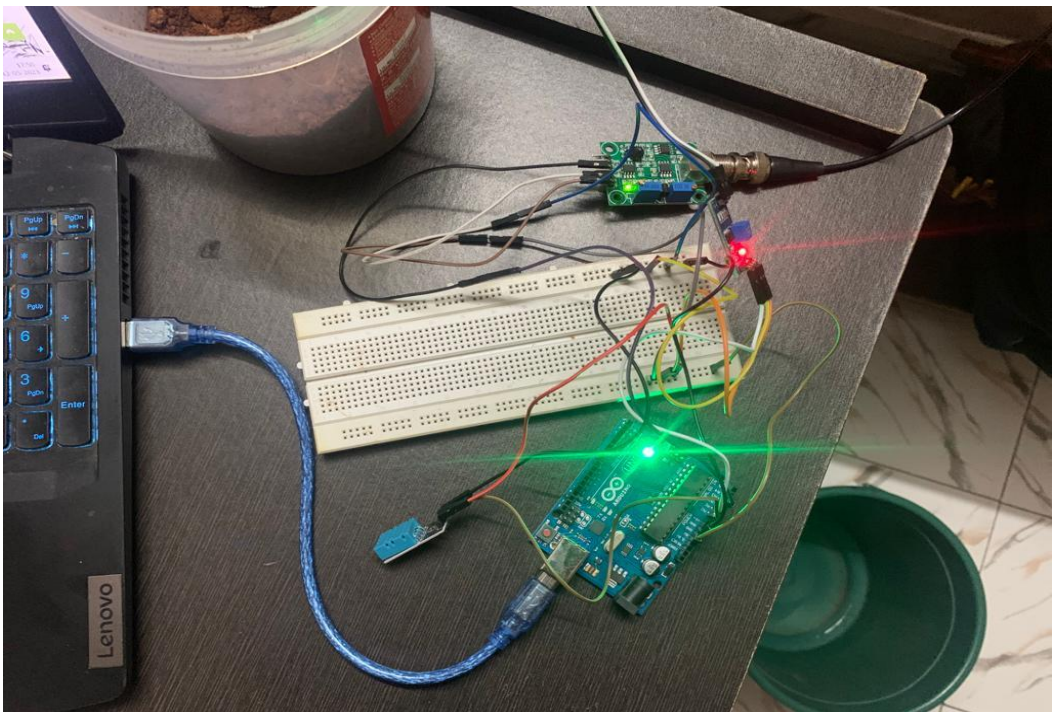
Prime Minister Narendra Modi launched a pension scheme for the small & marginal farmers of India last September. Under PM Kisan Maandhan scheme about 5 crore marginalised farmers will get a minimum pension of Rs 3000 / month on attaining the age of 60. Those who fall in the age group of 18 - 40 years will be eligible to apply for the scheme. Under this scheme, the farmers will be required to make a monthly contribution of Rs 55 to 200, depending on their age of entry, in the Pension Fund till they reach the retirement date, 60 years. The Government will make an equal contribution of the same amount in the pension fund for the cultivators.

For more details Click Here!

System Setup



Circuit Setup



Cloud Store View

1 Result | 2 Messages | 3 Table Data | 4 Objects | 5 History

Show All Or Limit 0 50 Refresh

Did	N_val	P_val	K_val	Temp_val	Humid_val	Ph_val	Moist_val	Prediction
1	90	33	86	33.00	74.00	4.12	98.700000000000003	mothbeans
2	23	42	86	33.00	77.00	4.14	99.9	papaya
(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)

Chapter 7

Testing and Results

7.1 Test Plan

7.1.1 Introduction

- Purpose: The purpose of this test plan is to ensure that the login, registration authentication, market analysis, crop prediction functions work correctly and provide an accurate analysis and prediction.
- Scope: The scope of this test plan is to test the login, registration authentication, market analysis, crop prediction, weather forecasting functions of the website.
- Test Environment: The testing will be conducted in a staging environment that simulates the production environment.
- Test Team: The testing will be conducted by the QA team.

7.1.2 Test Strategy:

Software testing methods are traditionally divided into black box testing and white box testing. These two approaches are used to describe the point of view that a test engineer takes when designing test cases.

7.2 Test Cases

Sr no.	Action	Inputs	Expected Output	Actual Output	Result	Comments
1	Collecting temperature value from DHT11	Air, Water, Soil	Between 20ºC-25 ºC	24- Air 25- Water 29- Soil	PASS	Temperature remains close to room temperature in all cases.
2	Collecting humidity value from DHT11	Air, Water, Soil	Between 50%-60%	54%- Air 54%- Water 60%- Soil	PASS	Humidity value remains the same in air and water but goes a little higher in soil.

3	Collecting sensor value from Soil moisture sensor	Air, Water, Soil	Decreases as we go from air to soil	1023- Air 400-450– Water 300-350- Soil	PASS	Sensor value ranges from 0-1023. It is maximum in air and decreases as we go from water to soil.
4	Collecting moisture value	Air, water, soil	0 in air >0 in soil and water	0%- Air 50%-60%- Water 30%-40%- Soil	PASS	Thus, as moisture increases, sensor value decreases. Moisture is minimum(0) in air and rises up to 60% in water.

7.3 Testing Methods Used:

7.3.1 Black box testing

Black box testing treats the software as a "black box"—without any knowledge of internal implementation. Black box testing methods include: equivalence partitioning, boundary value analysis, all-pairs testing, fuzz testing, model-based testing, traceability matrix, exploratory testing and specification-based testing.

- **Specification-based testing:**
  Specification-based testing aims to test the functionality of software according to the applicable requirements. Thus, the tester inputs data into, and only sees the output from, the test object. This level of testing usually requires thorough test cases to be provided to the tester, who then can simply verify that for a given input, the output value (or behavior), either "is" or "is not" the same as the expected value specified in the test case. Specification-based testing is necessary, but it is insufficient to guard against certain risks.

- **Advantages and disadvantages:**
  The black box tester has no "bonds" with the code, and a tester's perception is very simple: a code must have bugs. Using the principle, "Ask and you shall receive," black box testers find bugs where programmers do not. But, on the other hand, black box testing has been said to be "like a walk in a dark labyrinth without a flashlight," because the tester doesn't know how the software being tested was actually constructed. As a result, there are situations when (1) a tester writes many test cases to check something that could have been tested by only one test case, and/or (2) some parts of the back-end are not tested at all.
  Therefore, black box testing has the advantage of "an unaffiliated opinion," on the one hand, and the disadvantage of "blind exploring," on the other.

7.3.2 White box testing

White box testing is when the tester has access to the internal data structures and algorithms including the code that implement these.
Types of white box testing:
The following types of white box testing exist:

- **API testing (Application Programming Interface)** - Testing of the application using Public and Private APIs
- **Code coverage** - creating tests to satisfy some criteria of code coverage (e.g., the test designer can create tests to cause all statements in the program to be executed at least once)
- **Fault injection methods** - improving the coverage of a test by introducing faults to test code paths
- **Mutation testing methods**
  1. **Static testing** - White box testing includes all static testing
  2. **Code completeness evaluation**
     White box testing methods can also be used to evaluate the completeness of a test suite that was created with black box testing methods. This allows the software team to examine parts of a system that are rarely tested and ensures that the most important function points have been tested.
     Two common forms of code coverage are:
     - Function coverage, which reports on functions executed
     - Statement coverage, which reports on the number of lines executed to complete the test

     They both return code coverage metric, measured as a percentage.

7.3.3 Integration testing

Integration testing is any type of software testing, which seeks to verify the interfaces between components against a software design. Software components may be integrated in an iterative way or all together ("big bang"). Normally the former is considered a better practice since it allows interface issues to be localized more quickly and fixed.

7.3.4 Acceptance testing

Acceptance testing can mean one of two things:

1. A smoke test is used as an acceptance test prior to introducing a new build to the main testing process, i.e. before integration or regression.
2. Acceptance testing performed by the customer, often in their lab environment on their own HW, is known as user acceptance testing (UAT).

Conclusion

The crop recommendation system recommends crops to the consumer based at the preceding information and the present day geographic and climatic parameters of the consumer. Based at the input, an appropriate crop could be recommended. As of now, it's far constructed with most effective classifier algorithms and association rule which allows the farmers in deciding on the appropriate crop for his or her agriculture land through the proposed recommender model which in turn will increase the yield and the profit. Thus, in our proposed system, we have detailed an innovative approach for smart agriculture using two emerging technologies: Internet of Things and Machine Learning. With the use of both live and historical data, we ensure the accuracy of the result. Also comparing multiple ML algorithms enhances the accuracy of the system. Thus system will be used to reduce the difficulties faced by the farmers and will increase the quantity and quality of work done by them.

Future Scope

The use of IoT based crop yield prediction using a neural network has great potential for future development and implementation in the agriculture industry. Here are some potential future scopes for this technology:

1. Integration with precision farming technologies: As precision farming technologies continue to advance, there will be an increasing need for accurate crop yield prediction to optimize the use of resources such as water, fertilizer, and pesticides. The integration of IoT based crop yield prediction with precision farming technologies can help farmers make data-driven decisions and increase their crop yields.

2. Use of drone and satellite data: The use of drone and satellite data can provide valuable insights into the health and growth of crops. Combining IoT data with drone and satellite data can further improve crop yield prediction accuracy and provide more detailed information about crop health and potential yield.

3. Integration with blockchain technology: The integration of IoT based crop yield prediction with blockchain technology can help to create a more transparent and secure system for tracking and verifying crop yields. This can help to prevent fraud and ensure that farmers receive fair compensation for their crops.

4. Use of edge computing: Edge computing is a technology that allows data processing to occur closer to the source of the data, rather than in a centralized location. The use of edge computing in IoT based crop yield prediction can help to reduce latency and increase real-time data processing capabilities.

5. Implementation in developing countries: Developing countries often face challenges related to food security and agriculture, and IoT based crop yield prediction can provide valuable insights and solutions for these challenges. The future scope for this technology could involve implementing it in developing countries to improve crop yields and increase food security.

6. Use of multi-sensor data fusion: The combination of data from multiple sensors, such as weather sensors, soil moisture sensors, and IoT sensors, can provide a more comprehensive picture of crop health and yield potential. The future scope for IoT based crop yield prediction could involve exploring the use of multi-sensor data fusion to improve accuracy and provide more detailed insights into crop health.

   In conclusion, the future scope for IoT based crop yield prediction using a neural network is vast and has the potential to transform the agriculture industry. By integrating with other technologies and exploring new applications, this technology can help to optimize resource allocation, increase crop yields, and improve food security.

References

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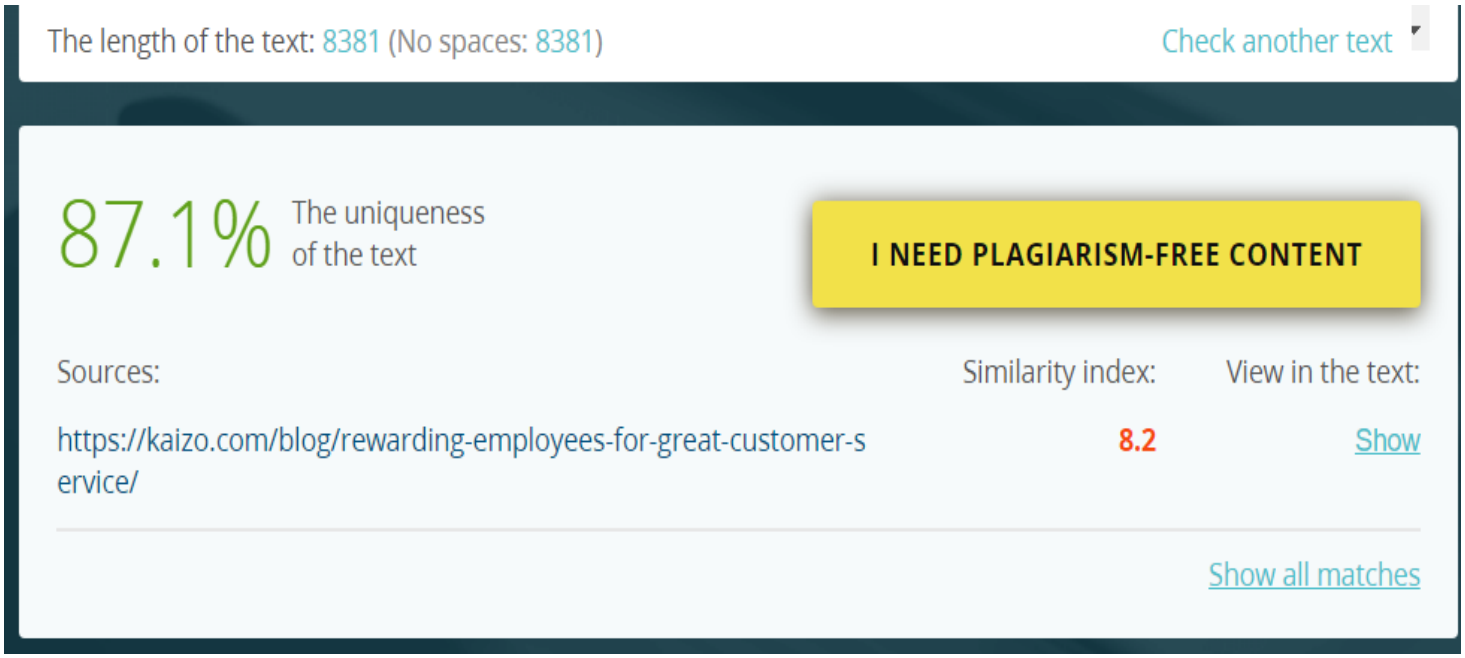
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